



Introduction to Python for Finance

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Course Description

Created in 1991 by Guido van Rossum, Python has become one of the favorite programming languages both in academia and in industry, thanks to its simple syntax and extensive (powerful) libraries. The course seeks to provide the basic tools to tackle financial and/or economic problems. Among the topics to be discussed are: data structures, conditional control flow, functions, an introduction to the NumPy library, data manipulation with Pandas and time series.

Objectives

1. Understand Python's syntax and programming environment
2. Become familiar with the different data types and data structures
3. Manipulate structured datasets using Pandas
4. Create your own functions to solve real-world problems
5. Present results using Jupyter Notebooks

Grading

The final grade will be weighted as follows:

- Homework: 70% (maximum 2 members per group)
- Final Exam: 30% (covers all the course material)

Methodology

The course will be taught through lectures and hands-on exercises using Python.

Contents

1. Introduction
 - a. Installation and Setup of Anaconda (IDEs)
 - b. The Python Interpreter
 - c. IPython Shell
 - d. Jupyter Notebook
 - e. Google Colaboratory
2. Data Structures & Sequences
 - a. Tuple (tuple)
 - b. List (list)
 - c. Dictionary (dict)
 - d. Set (set)
3. Conditional Control Flow
 - a. for and while Loops
 - b. if, elif and else
 - c. Logical Operators
 - d. Mathematical Operators
 - e. List, Set and Dictionary Comprehension
4. Functions
 - a. Creating and Importing
 - b. Itertools Module
 - c. Anonymous Functions (Lambda)
 - d. Error and Exception Handling
 - e. Importing Libraries
 - f. Example: Bullet Bond Price
5. Introduction to NumPy
 - a. Array, Matrix and ndarray
 - b. Concatenation, Splitting and Subsetting
 - c. Array Computation: Broadcasting
 - d. Linear Algebra Functions
 - e. Example: Linear Regression
 - f. Example: *Random Walk*

6. Data Manipulation with Pandas
 - a. DataFrame, Series and Index Object
 - b. Essential Functions for Pandas Data Structures
 - c. Handling *Missing Values* and *Null Values*
 - d. Variable Transformation
 - e. Data Aggregation and Group Operations
 - f. Combining Datasets: Concat, Append, Merge and Join
 - g. Reshape and Pivot
 - h. Importing and Exporting Data
 - i. Example: Generating *Dummies*
7. Time Series
 - a. Time Series Data Structures in Pandas
 - b. *Resampling* and *Shifting* (Building Returns)
 - c. *Rolling Window* and *Expanding Window*
 - d. Example: *Equally Weighted* Portfolio
 - e. Example: Geometric Cumulative Return

Recommended Readings

McKinney, Wes (2017). *Python for data analysis: Data wrangling with Pandas, NumPy, and IPython*. O'Reilly Media, Inc.

Ramalho, Luciano (2015). *Fluent python: Clear, concise, and effective programming*. O'Reilly Media, Inc.

Sheppard, Kevin (2019). "Introduction to Python for econometrics, statistics and data analysis". In: *Self-published, University of Oxford, version 3*.

VanderPlas, Jake (2016). *Python data science handbook: Essential tools for working with data*. O'Reilly Media, Inc.